

DeepSeek-V2: Efficient MoE Language Model with Multi-Head Latent Attention

Cost-performance frontier for long-context LLMs

Scaling Dilemma

- Better quality needs larger models and longer context
- Inference bottleneck shifts to KV cache memory
- Dense scaling raises training and serving costs
- Goal: move the Pareto frontier

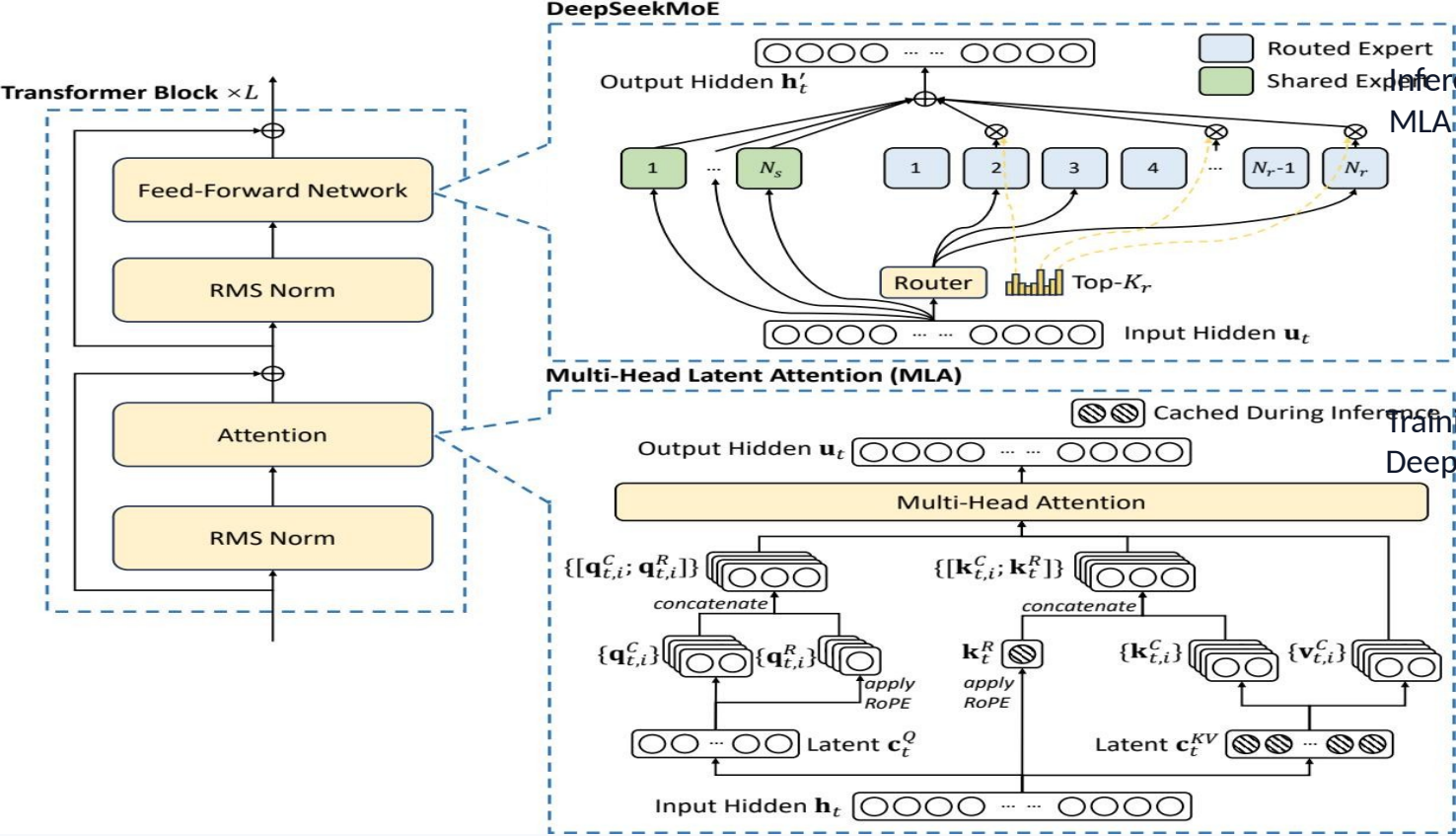
DeepSeek-V2 at a glance

236B total params | 21B activated/token
128K context | 8.1T pretraining tokens
SFT (1.5M sessions) + GRPO RL

Core innovations: MLA + DeepSeekMoE

Architecture Overview: MLA + DeepSeekMoE in a 236B/21B Sparse Stack

Two-pronged design for inference and training efficiency



Inference path

MLA compresses KV state into latent cache and reconstructs on demand.

Training path

DeepSeekMoE sparsely activates specialized experts to cut compute cost

Why Dense Scaling Breaks: Memory-Bound Inference and Rising Training Spend

Problem framing before architectural interventions

Dense LLM bottlenecks

- KV cache grows with sequence length
- Throughput drops as memory traffic dominates
- Training cost scales aggressively at high quality targets
- Long context magnifies serving inefficiency

Conventional attention trade-offs

MHA: best baseline quality, largest KV cache

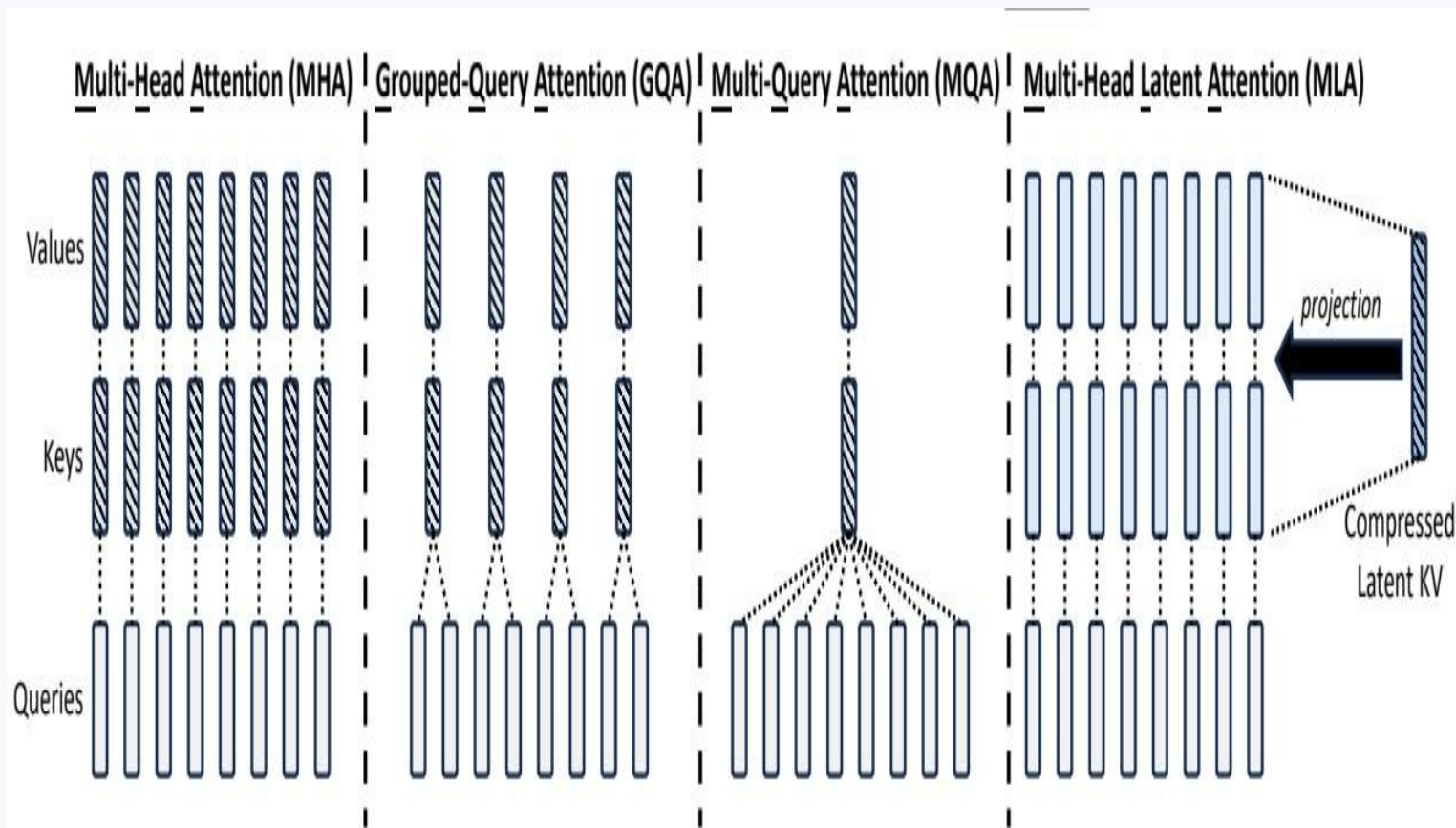
GQA: smaller KV cache, possible quality trade-off

MQA: strongest KV reduction, larger quality risk

DeepSeek-V2 objective: preserve quality while collapsing memory+compute overhead.

Multi-Head Latent Attention (MLA): Low-Rank KV Compression without Quality Loss

Cache compact latent states, reconstruct full representations on-the-fly



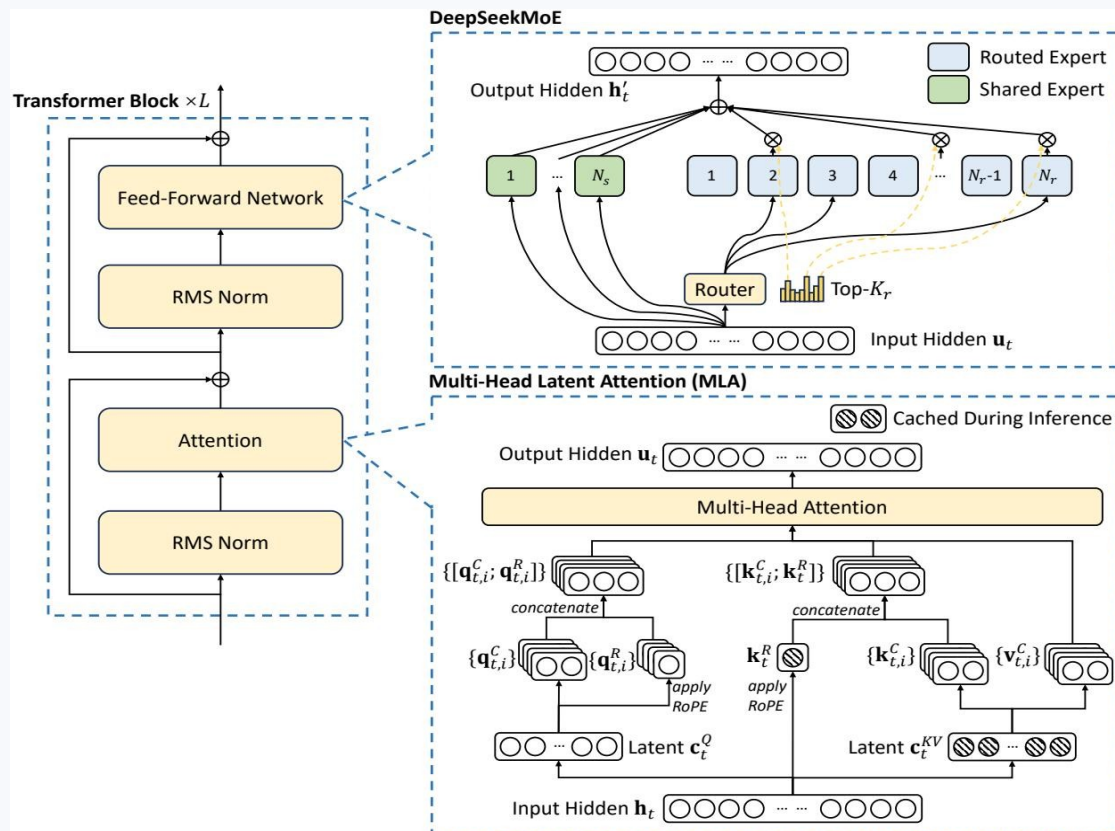
MLA mechanism

- 1) Project K/V into low-rank latent c_t^{KV}
- 2) Cache latent vector instead of full per-head K/V
- 3) Reconstruct keys/values using learned projections
- 4) Use decoupled RoPE positional key

Outcome: better than MHA with much smaller cache.

DeepSeekMoE: Fine-Grained Experts with Shared Expert Isolation

Sparse FFN design for specialization and communication efficiency



Mechanics

- Shared experts (N_s): always active for common knowledge
- Routed experts (N_r): top- K_r token-wise specialization
- Device-limited routing: at most M devices/token
- Auxiliary losses balance expert and device utilization
- Training-time token dropping avoids overflow

Net effect: high capacity at lower effective training compute.

5% Training Cost Reduction and 5.76× Generation Throughput vs. DeepSeek 67B

Relative metrics from source brief only

DeepSeek 67B (Baseline)
Reference cost, cache, and throughput

DeepSeek-V2
42.5% lower training cost
93.3% lower KV cache
5.76× higher max generation throughput

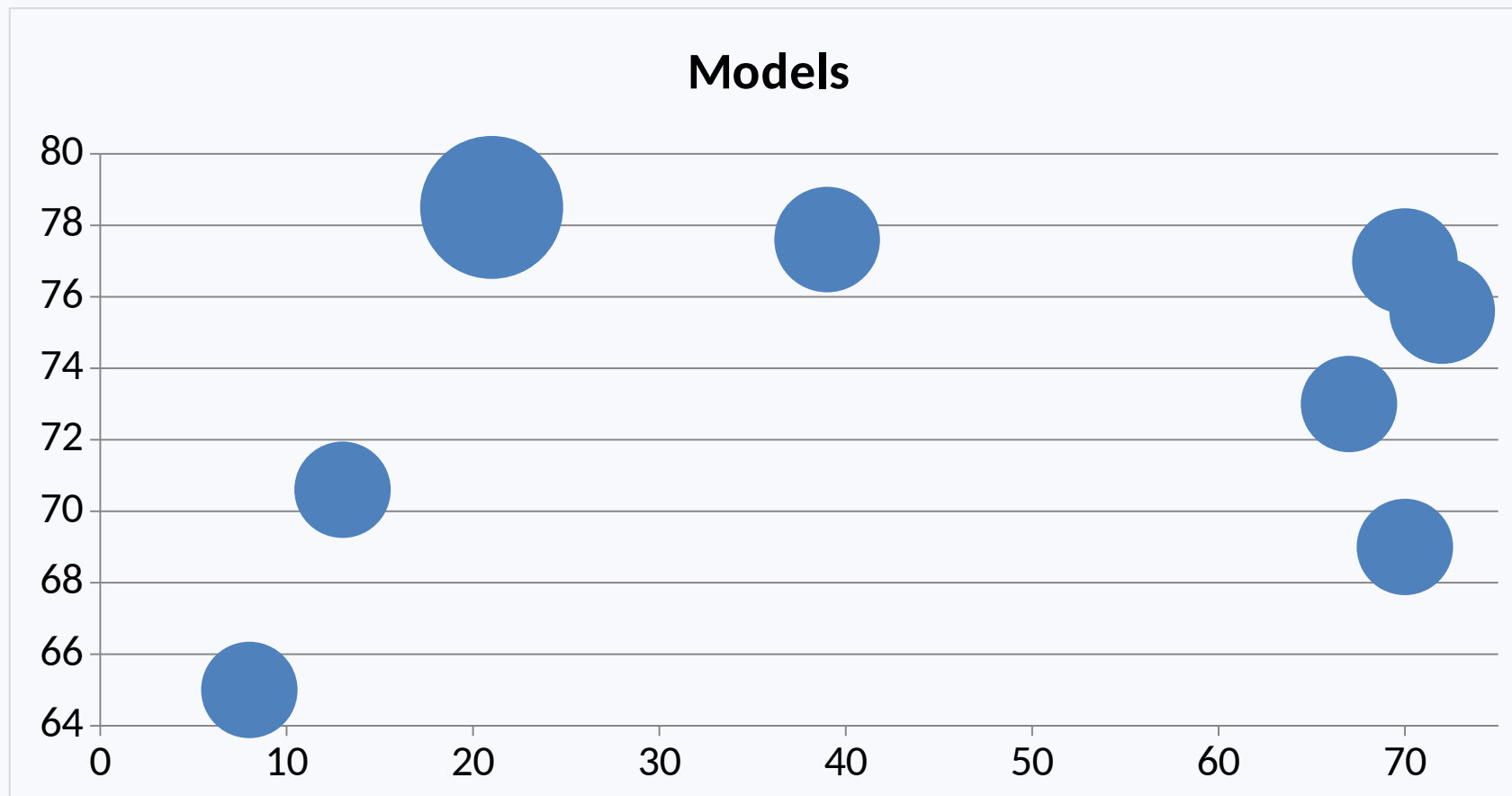
Training Cost
-42.5%

KV Cache Size
-93.3%

Max Gen Throughput
5.76×

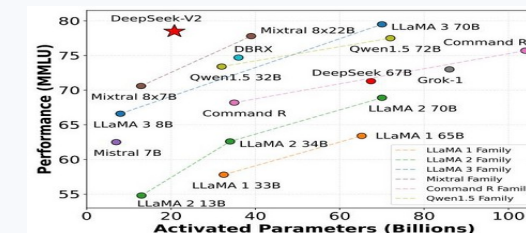
Grok-V2 Tops MMLU Among Open-Source Models at Only 21B Activated Parameters

MMLU vs activated parameters (scatter-style)



Pareto signal

- 78.5 MMLU at 21B activated params
- Beats LLaMA 3 70B (77.0)
- Beats Qwen1.5 72B (75.6)
- Strong quality-efficiency frontier position



DeepSeek-V2 Chat (RL) Rivals Closed-Source Models on Chinese Benchmarks

Alignment pipeline and chat evaluation outcomes

Alignment pipeline

SFT: 1.5M conversational sessions

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RL: Group Relative Policy Optimization (GRPO)

Outcome: improved open-ended quality and multilingual alignment, especially Chinese

Chat evaluation (DeepSeek-V2 Chat RL)

AlpacaEval 2.0 win rate: 38.9%

MT-Bench (overall): 8.97

AlignBench (overall, Chinese): 7.91

AlignBench: outperforms all open-source and most closed-source models in Chinese.

Key Takeaways: Architectural Innovation Unlocks Efficient Scaling

DeepSeek-V2 as a Pareto-optimal open-source design point

- MLA provides superior KV-cache efficiency while maintaining strong attention quality
- DeepSeekMoE enables economical scaling with shared+routed expert structure
- Relative to DeepSeek 67B: 42.5% lower training cost and 93.3% lower KV cache
- 5.76× higher generation throughput improves deployment practicality
- 21B activated parameters still deliver top-tier MMLU and robust alignment